DBMS LAB – SQL COMMANDS

Each question is followed by separate query and output screenshots.

Name:DIVYANTH T

Reg no:1912260710

Q1. DDL Commands using CREATE, ALTER, TRUNCATE and DROP

Screenshot:

```sql
CREATE DATABASE IF NOT EXISTS dbms_lab;
USE dbms_lab;

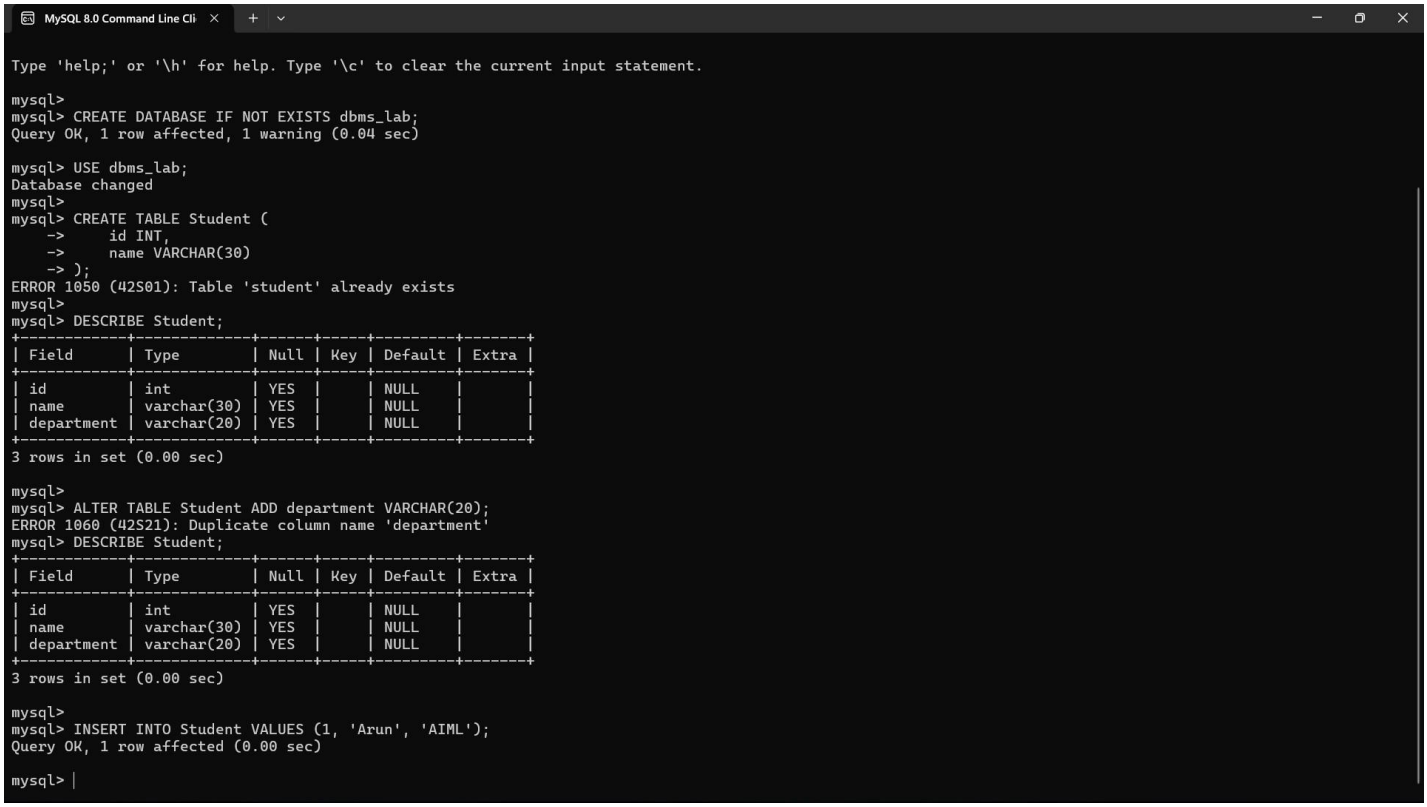
CREATE TABLE Student (
    id INT,
    name VARCHAR(30)
);

DESCRIBE Student;

ALTER TABLE Student ADD department VARCHAR(20);
DESCRIBE Student;

INSERT INTO Student VALUES (1, 'Arun', 'AIML');
```

Output Screenshot


MySQL 8.0 Command Line Cli
Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.
mysql>
mysql> CREATE DATABASE IF NOT EXISTS dbms_lab;
Query OK, 1 row affected, 1 warning (0.04 sec)
mysql> USE dbms_lab;
Database changed
mysql>
mysql> CREATE TABLE Student (
-> id INT,
-> name VARCHAR(30)
->);
ERROR 1050 (42S01): Table 'student' already exists
mysql>
mysql> DESCRIBE Student;
+------------+-------------+------+-----+---------+-------+
| Field | Type | Null | Key | Default | Extra |
+------------+-------------+------+-----+---------+-------+
id	int	YES		NULL	
name	varchar(30)	YES		NULL	
department	varchar(20)	YES		NULL	
+------------+-------------+------+-----+---------+-------+
3 rows in set (0.00 sec)
mysql>
mysql> ALTER TABLE Student ADD department VARCHAR(20);
ERROR 1060 (42S21): Duplicate column name 'department'
mysql> DESCRIBE Student;
+------------+-------------+------+-----+---------+-------+
| Field | Type | Null | Key | Default | Extra |
+------------+-------------+------+-----+---------+-------+
id	int	YES		NULL	
name	varchar(30)	YES		NULL	
department	varchar(20)	YES		NULL	
+------------+-------------+------+-----+---------+-------+
3 rows in set (0.00 sec)
mysql>
mysql> INSERT INTO Student VALUES (1, 'Arun', 'AIML');
Query OK, 1 row affected (0.00 sec)
mysql>

Q2. DDL Commands with Constraints

Screenshot:

```sql
CREATE TABLE Department (
    dept_id INT PRIMARY KEY,
    dept_name VARCHAR(30) UNIQUE
);

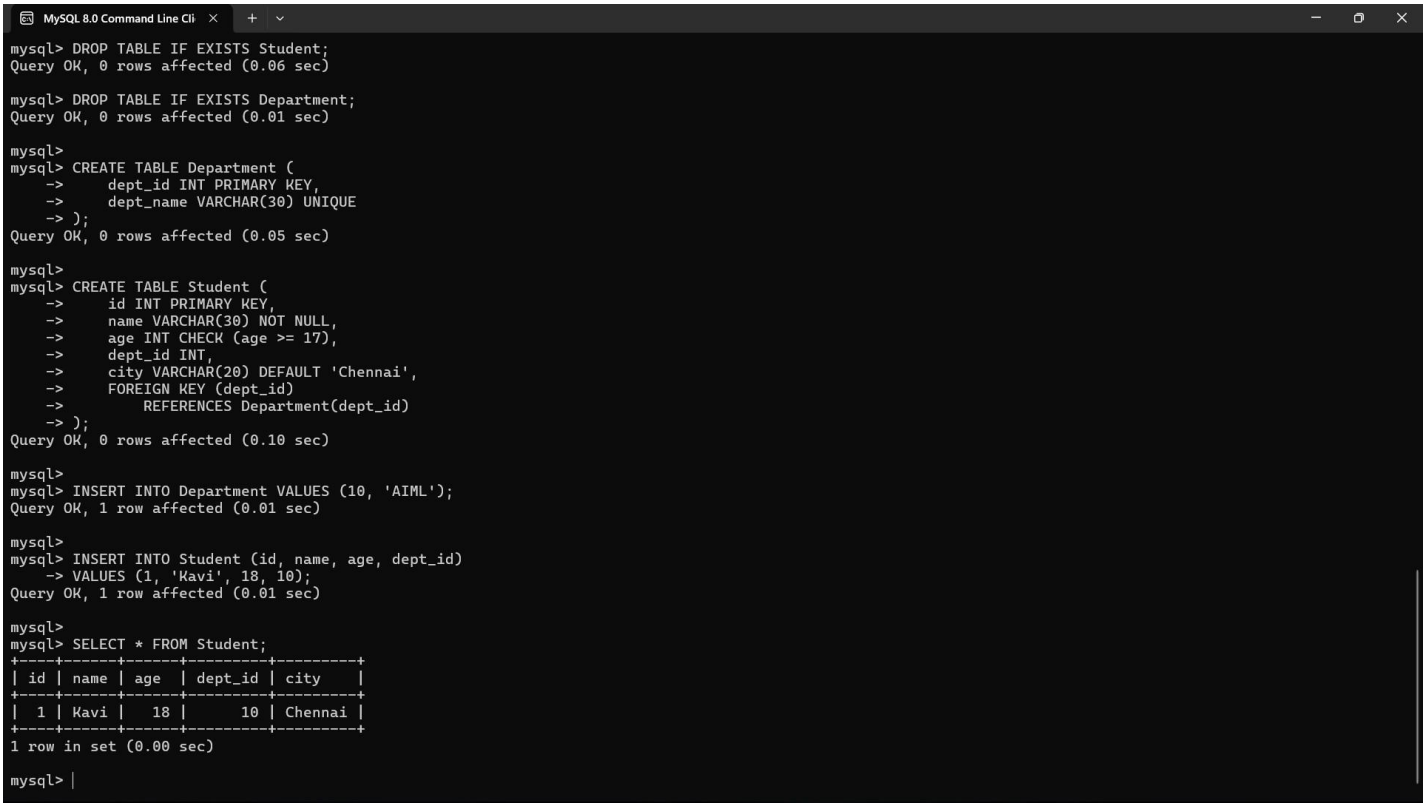
CREATE TABLE Student (
    id INT PRIMARY KEY,
    name VARCHAR(30) NOT NULL,
    age INT CHECK (age >= 17),
    dept_id INT,
    city VARCHAR(20) DEFAULT 'Chennai',
    FOREIGN KEY (dept_id)
        REFERENCES Department(dept_id)
);

INSERT INTO Department VALUES (10, 'AIML');

INSERT INTO Student (id, name, age, dept_id)
VALUES (1, 'Kavi', 18, 10);

SELECT * FROM Student;
```

Output screenshot:


MySQL 8.0 Command Line Cli
mysql> DROP TABLE IF EXISTS Student;
Query OK, 0 rows affected (0.06 sec)
mysql> DROP TABLE IF EXISTS Department;
Query OK, 0 rows affected (0.01 sec)
mysql>
mysql> CREATE TABLE Department (
-> dept_id INT PRIMARY KEY,
-> dept_name VARCHAR(30) UNIQUE
->);
Query OK, 0 rows affected (0.05 sec)
mysql>
mysql> CREATE TABLE Student (
-> id INT PRIMARY KEY,
-> name VARCHAR(30) NOT NULL,
-> age INT CHECK (age >= 17),
-> dept_id INT,
-> city VARCHAR(20) DEFAULT 'Chennai',
-> FOREIGN KEY (dept_id)
-> REFERENCES Department(dept_id)
->);
Query OK, 0 rows affected (0.10 sec)
mysql>
mysql> INSERT INTO Department VALUES (10, 'AIML');
Query OK, 1 row affected (0.01 sec)
mysql>
mysql> INSERT INTO Student (id, name, age, dept_id)
-> VALUES (1, 'Kavi', 18, 10);
Query OK, 1 row affected (0.01 sec)
mysql>
mysql> SELECT * FROM Student;
+----+------+------+---------+---------+
| id | name | age | dept_id | city |
+----+------+------+---------+---------+
| 1 | Kavi | 18 | 10 | Chennai |
+----+------+------+---------+---------+
1 row in set (0.00 sec)
mysql>

Q3. DML Commands using INSERT and SELECT

Screenshot:

```sql
USE dbms_lab;

DROP TABLE IF EXISTS Student;

CREATE TABLE Student (
    id INT,
    name VARCHAR(30),
    course VARCHAR(20)
);

INSERT INTO Student VALUES (1, 'Anu', 'AIML');
INSERT INTO Student VALUES (2, 'Bala', 'CSE');

SELECT * FROM Student;
```

Output Screenshot:

```
mysql>
mysql> DROP TABLE IF EXISTS Student;
Query OK, 0 rows affected (0.06 sec)

mysql>
mysql> CREATE TABLE Student (
    ->     id INT,
    ->     name VARCHAR(30),
    ->     course VARCHAR(20)
    -> );
Query OK, 0 rows affected (0.02 sec)

mysql>
mysql> INSERT INTO Student VALUES (1, 'Anu', 'AIML');
Query OK, 1 row affected (0.01 sec)

mysql> INSERT INTO Student VALUES (2, 'Bala', 'CSE');
Query OK, 1 row affected (0.01 sec)

mysql>
mysql> SELECT * FROM Student;
+------+------+--------+
| id   | name | course |
+------+------+--------+
|    1 | Anu  | AIML   |
|    2 | Bala | CSE    |
+------+------+--------+
2 rows in set (0.00 sec)
```

Q4. DML Commands using UPDATE and DELETE

screenshot:

```sql
DROP TABLE IF EXISTS Student;

CREATE TABLE Student (
    id INT,
    name VARCHAR(30),
    marks INT
);

INSERT INTO Student VALUES (1, 'Anu', 75);
INSERT INTO Student VALUES (2, 'Bala', 68);

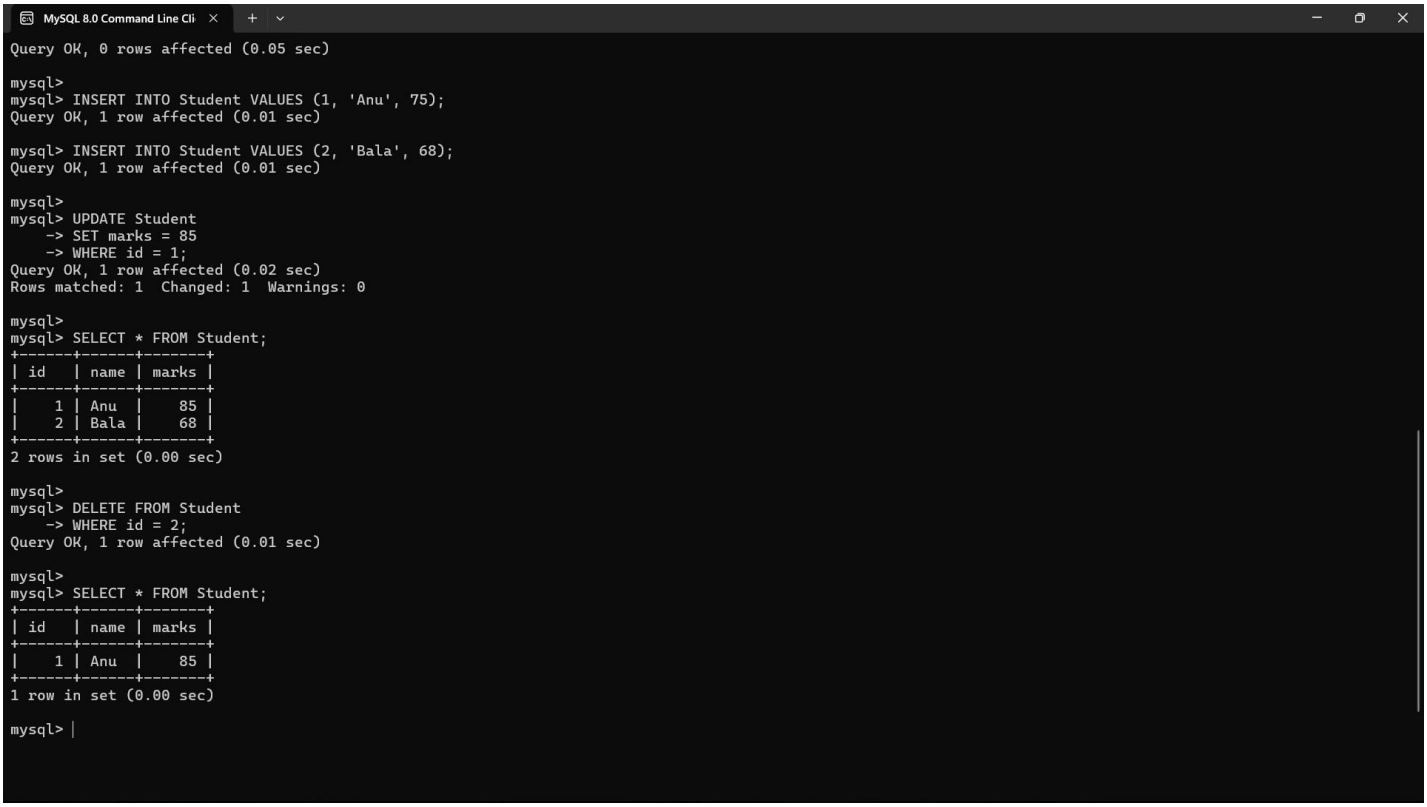
UPDATE Student
SET marks = 85
WHERE id = 1;

SELECT * FROM Student;

DELETE FROM Student
WHERE id = 2;

SELECT * FROM Student;
```

output Screenshot:



Q5. TCL Commands – COMMIT, SAVEPOINT and ROLLBACK

Screenshot:

1

```sql
USE dbms_lab;

DROP TABLE IF EXISTS Account;

CREATE TABLE Account (
    id INT PRIMARY KEY,
    balance INT
) ENGINE = InnoDB;

INSERT INTO Account VALUES (1, 1000);

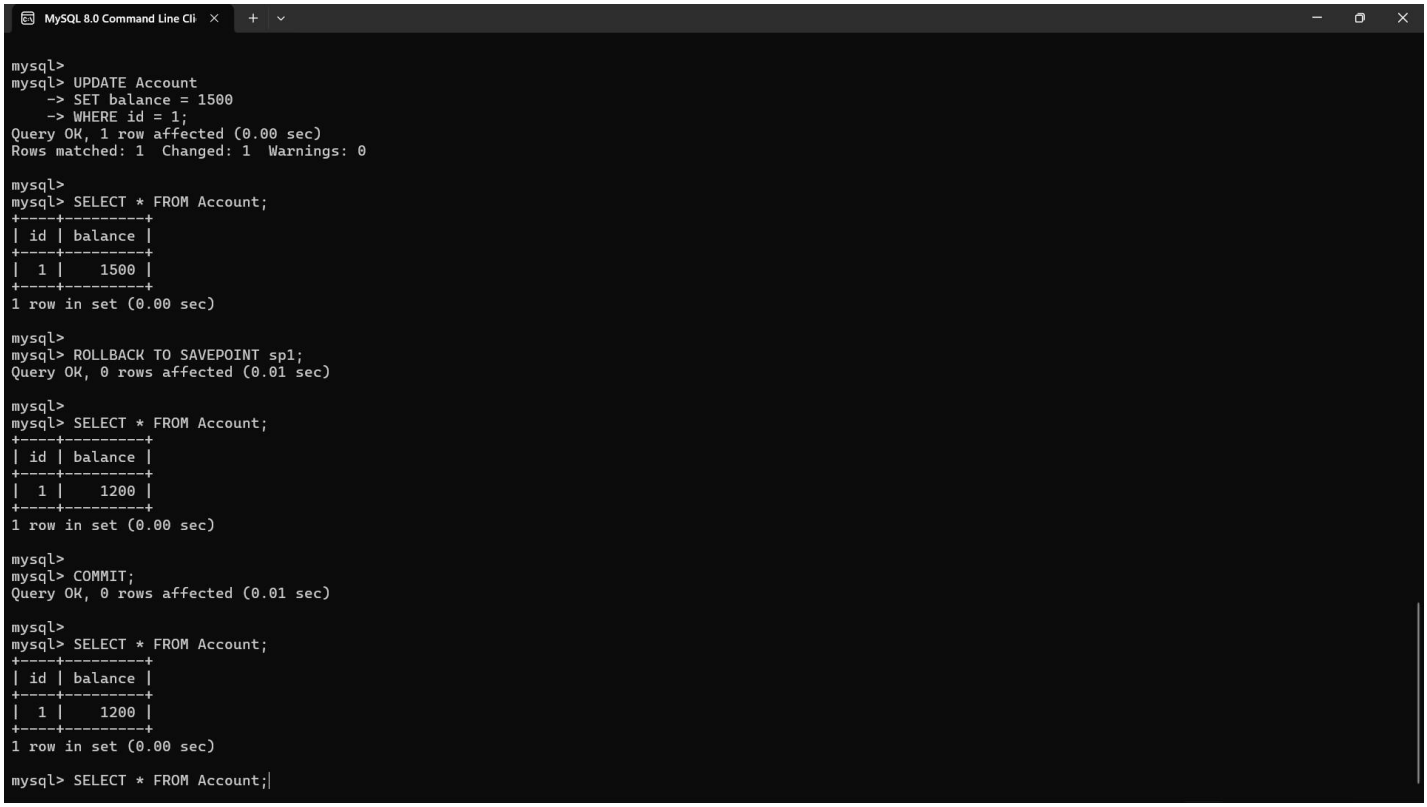
START TRANSACTION;

UPDATE Account
SET balance = 1200
WHERE id = 1;

SAVEPOINT sp1;

UPDATE Account
```

Output Screenshot:


MySQL 8.0 Command Line Cli
mysql>
mysql> UPDATE Account
-> SET balance = 1500
-> WHERE id = 1;
Query OK, 1 row affected (0.00 sec)
Rows matched: 1 Changed: 1 Warnings: 0
mysql>
mysql> SELECT * FROM Account;
+----+---------+
| id | balance |
+----+---------+
| 1 | 1500 |
+----+---------+
1 row in set (0.00 sec)
mysql>
mysql> ROLLBACK TO SAVEPOINT sp1;
Query OK, 0 rows affected (0.01 sec)
mysql>
mysql> SELECT * FROM Account;
+----+---------+
| id | balance |
+----+---------+
| 1 | 1200 |
+----+---------+
1 row in set (0.00 sec)
mysql>
mysql> COMMIT;
Query OK, 0 rows affected (0.01 sec)
mysql>
mysql> SELECT * FROM Account;
+----+---------+
| id | balance |
+----+---------+
| 1 | 1200 |
+----+---------+
1 row in set (0.00 sec)
mysql> SELECT * FROM Account;